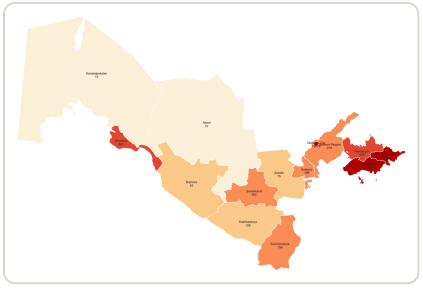
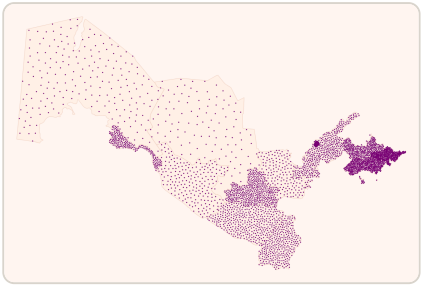
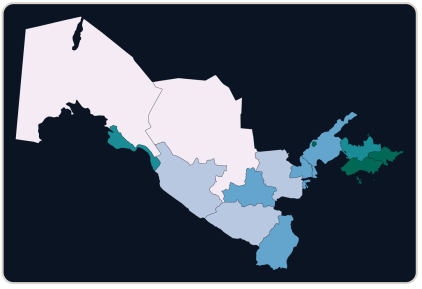
🟪 Day 03 · Polygons: how many people live in each region?

A beginner tutorial with Mapshaper, right in your browser. No account, no installation, no experience needed.

You will load the 14 regions of Uzbekistan as **polygons**, measure their **area**, attach the official **population** of each region, calculate the **population density** (people per km²) and colour the map by it. Then you will turn the same numbers into a **dot map** where every dot stands for thousands of people. On the way you will learn your first commands in a tiny map language.

⏱️ About 25 minutes · 💻 Any computer with a browser (Chrome, Edge, Firefox or Safari) · 🔑 No account needed

People per km²	One dot = 10,000 people	Night palette
		

📚 Part 0 · The ideas you need

Polygon. A closed shape that covers an area: a region, a lake, a park. Our file has one polygon for each of the 14 regions (12 regions, Karakalpakstan and Tashkent City).

Attribute table. Every polygon has a row of information next to its shape, like a spreadsheet. At the start there is only one column, `region`, with the name.

Join. Our population numbers are in a separate table (a CSV file). A *join* glues the two tables together by matching the region names.

Density. Big regions can have many people and still feel empty. Dividing people by area gives **people per km²**, which is fairer to compare.

Choropleth. A map where each polygon is coloured by its number. Darker = more people per km².

Mapshaper (mapshaper.org) is a free, open-source map tool that runs in your browser. Your files stay on your computer: nothing is uploaded to a server.

📥 Part 1 · Get the data

You need two small files. Download both into the same folder (for example *Downloads*):

File	What's inside
`uzbekistan_regions.geojson`	the 14 regions as polygons (column `region`)
`population_2026.csv`	the permanent population of each region on 1 January 2026 (columns `region`, `population`)

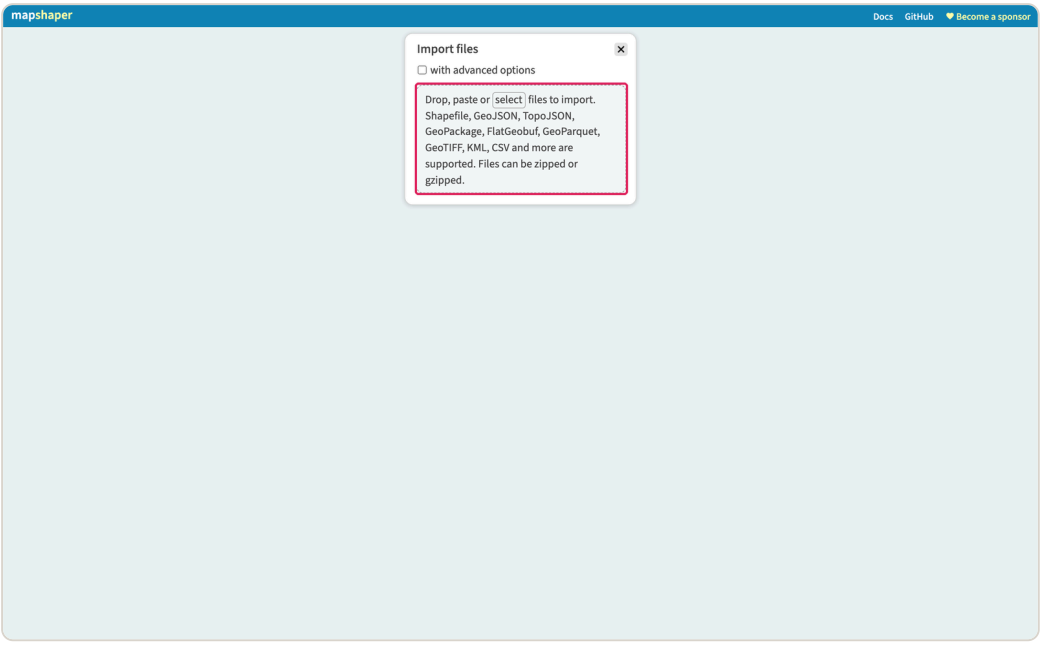
⬇️ Or download both at once: **day03_polygons_data.zip** and unzip it.

To download one file from GitHub: click its name, then click the **Download raw file** button (⬇) at the top right of the page.

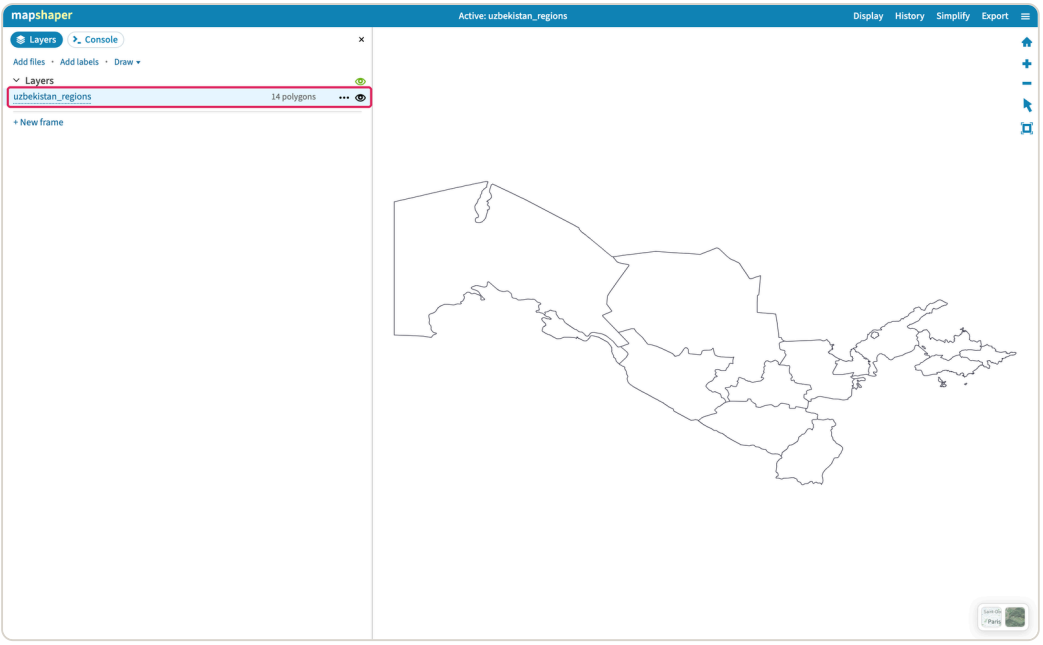
The population numbers are official figures from the **Statistics Agency of Uzbekistan** (stat.uz). Together they add up to the country total: **38,236,704 people**.

🗺️ Part 2 · Open Mapshaper and load the regions

1. Go to **mapshaper.org**. You see the **Import files** window.
2. Drag `uzbekistan_regions.geojson` into the dashed box (or click **select** and choose the file).



3. The regions appear on the right. On the left, the **Layers** list shows one layer, `uzbekistan_regions`, with **14 polygons**.

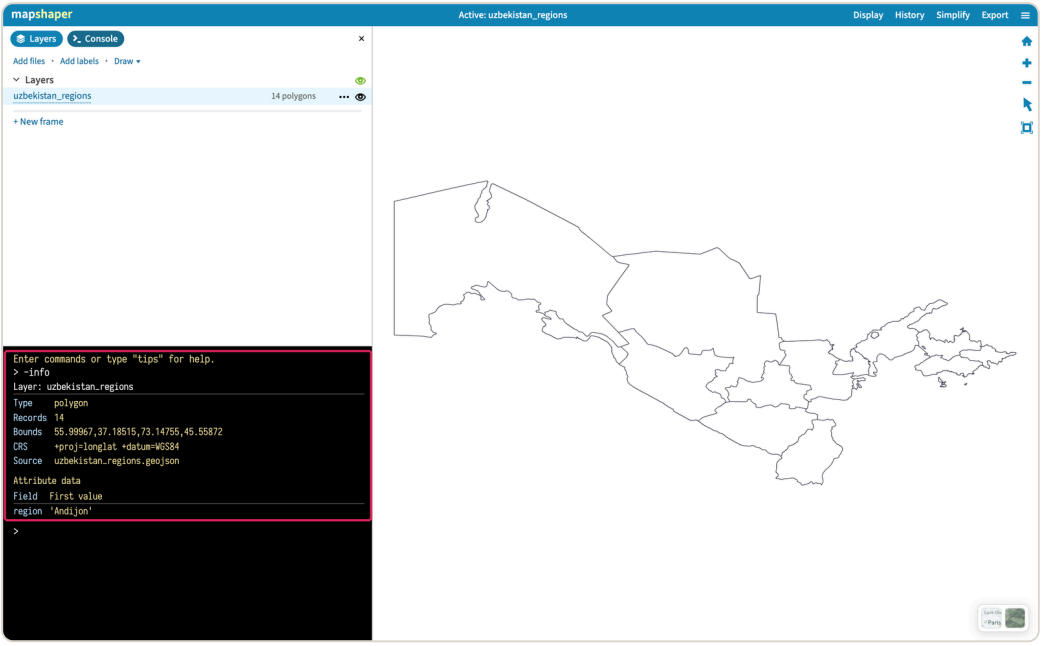


⌨️ Part 3 · Your first command

Click **Console** (top left). A black box opens at the bottom. Here you type **commands**: short instructions that start with a dash `–`.

Type this and press **Enter**:

```
–info
```



Mapshaper describes the layer: it is a **polygon** layer with **14 records** (rows), and its only field (column) is `region`.

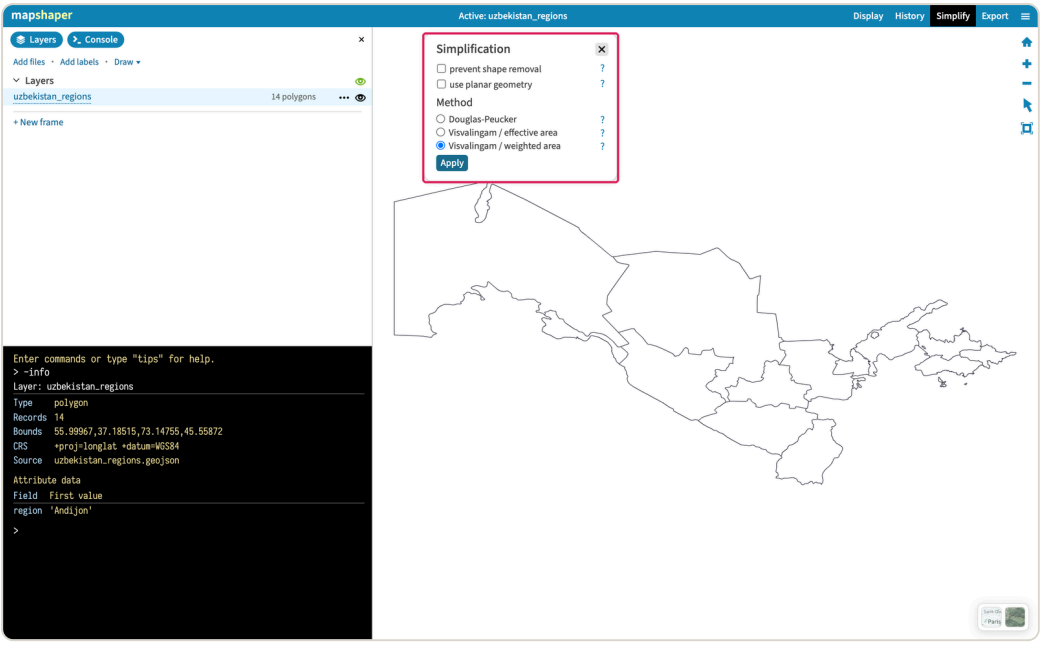
> 💡 You can type `tips` in the console for help, and use the ↑ arrow key to bring back the previous command. If you make a mistake, the ↶ **undo** button at the bottom of the map takes back the last

> change.

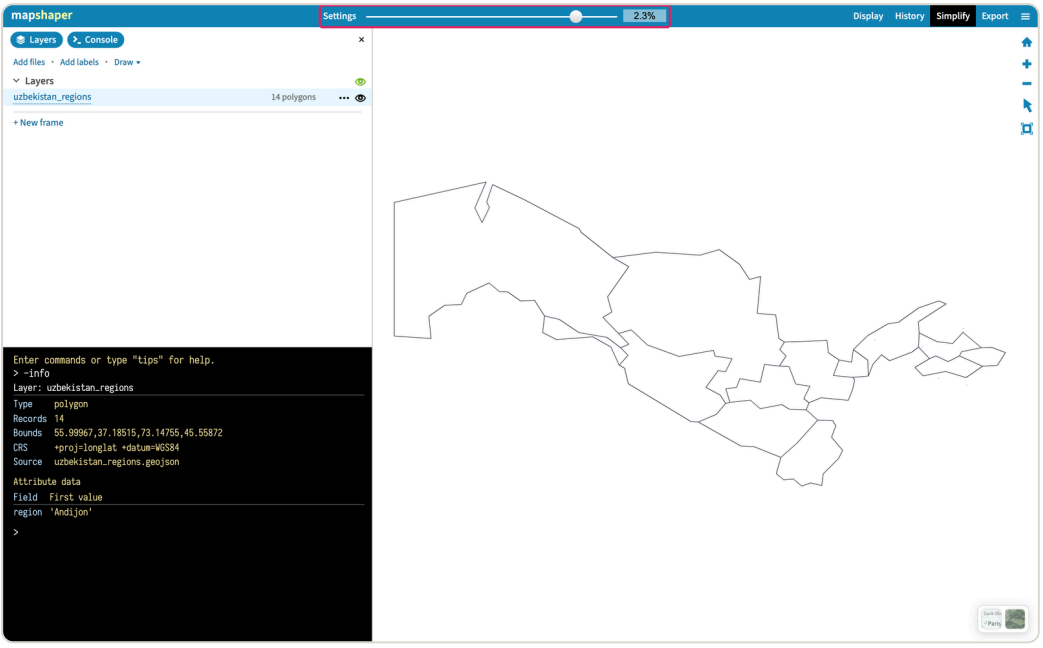
✂️ Part 4 · How detailed is a border? (Simplify)

Every border is made of thousands of little points. Let's see what happens with fewer of them.

1. Click **Simplify** (top right). Keep the default settings and click **Apply**.



2. A slider appears at the top. Drag it to the right, to about **3%**: only 3% of the points are kept and the borders become blocky.



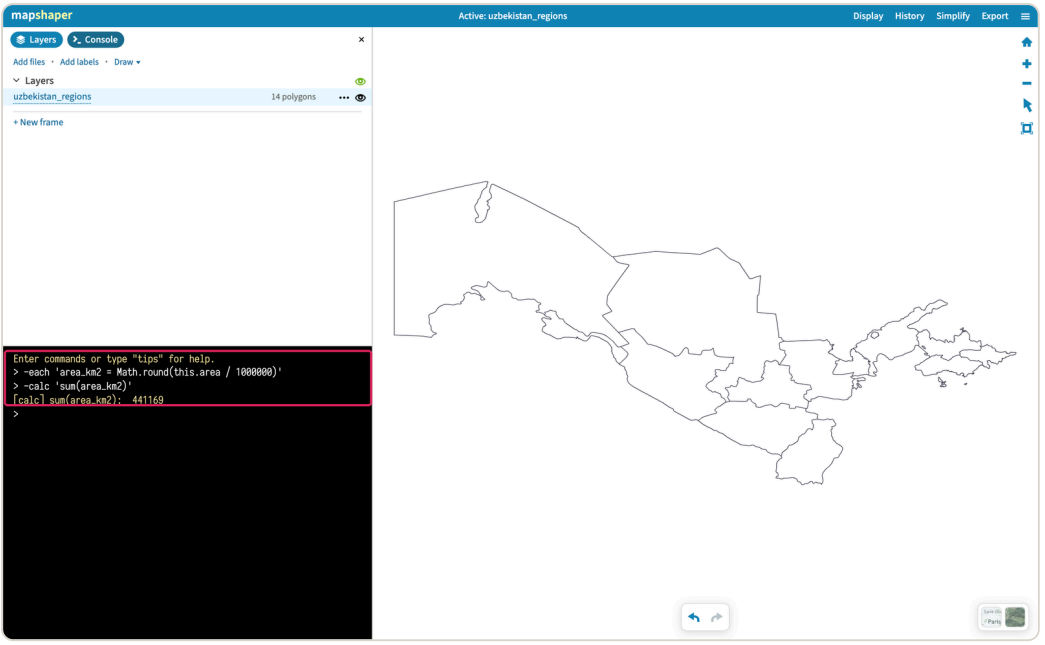
3. **Drag the slider all the way back to the left (100%)** before you continue, so your areas are measured on the full, detailed borders. Then click **Simplify** again to close it.

Simplifying makes files smaller and maps faster. For a whole-country map, 10–20% usually still looks great.

📐 Part 5 · Measure each polygon's area

Back in the console, type these two commands (press **Enter** after each one):

```
-each 'area_km2 = Math.round(this.area / 1000000)'
-calc 'sum(area_km2)'
```



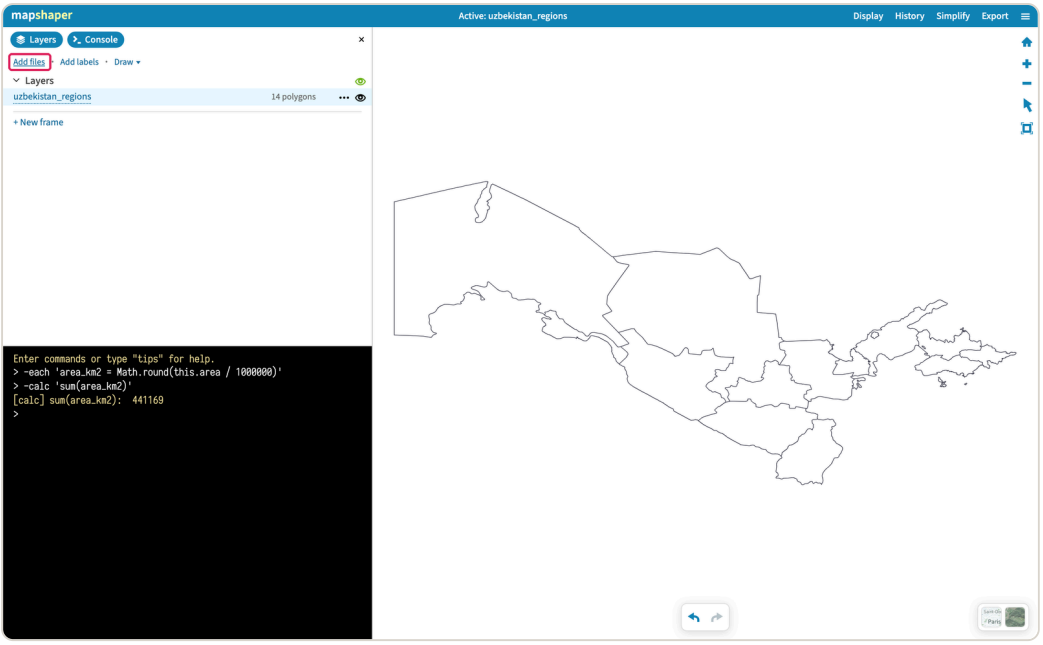
What happened:

- `-each` runs a small formula **for each polygon**. `this.area` is the polygon's area in m². Dividing by 1,000,000 gives km², and `Math.round` removes the decimals. The result goes into a new column, `area_km2`.
- `-calc 'sum(area_km2)'` adds up the column: about **441,000 km²** for the whole country.

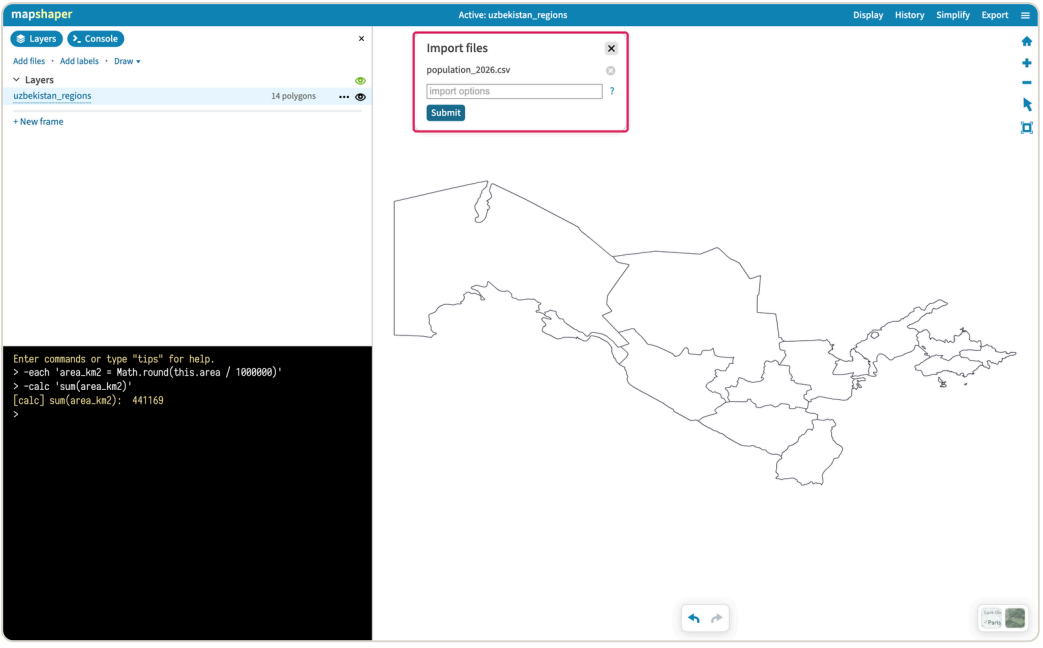
> ✏️ The quotes `' '` around the formula are important. Copy the commands exactly.

👥 Part 6 · Add the population table

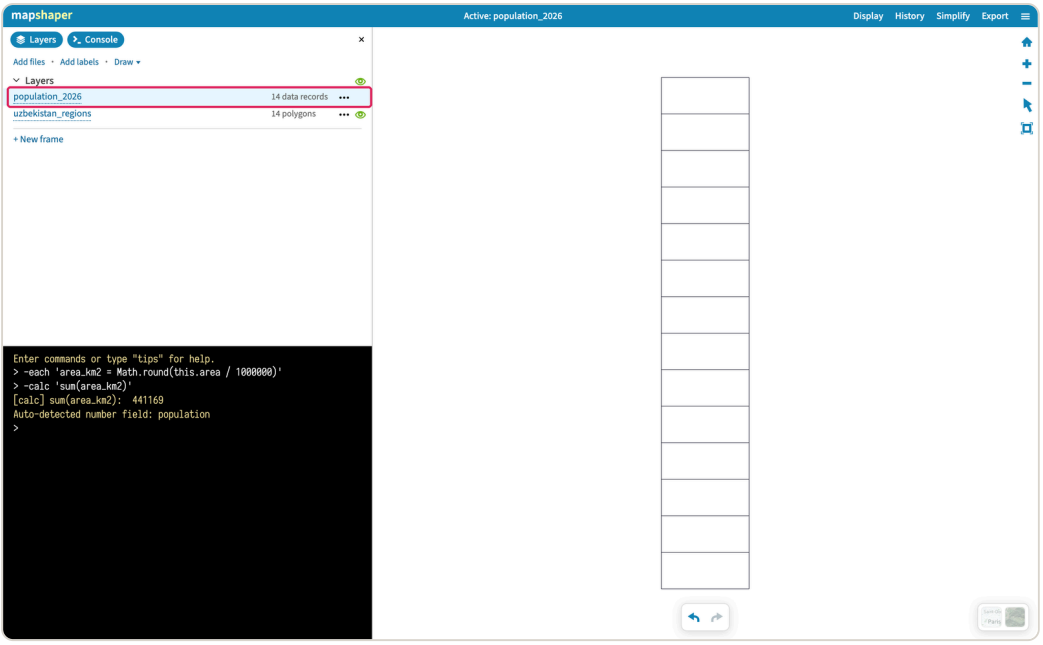
1. Click **Layers**, then **Add files** (above the layer list) and choose `population_2026.csv`.



2. Click **Submit**.



3. Now there are two layers: `population_2026` (**14 data records**: a table without shapes) and `uzbekistan_regions` (**14 polygons**).

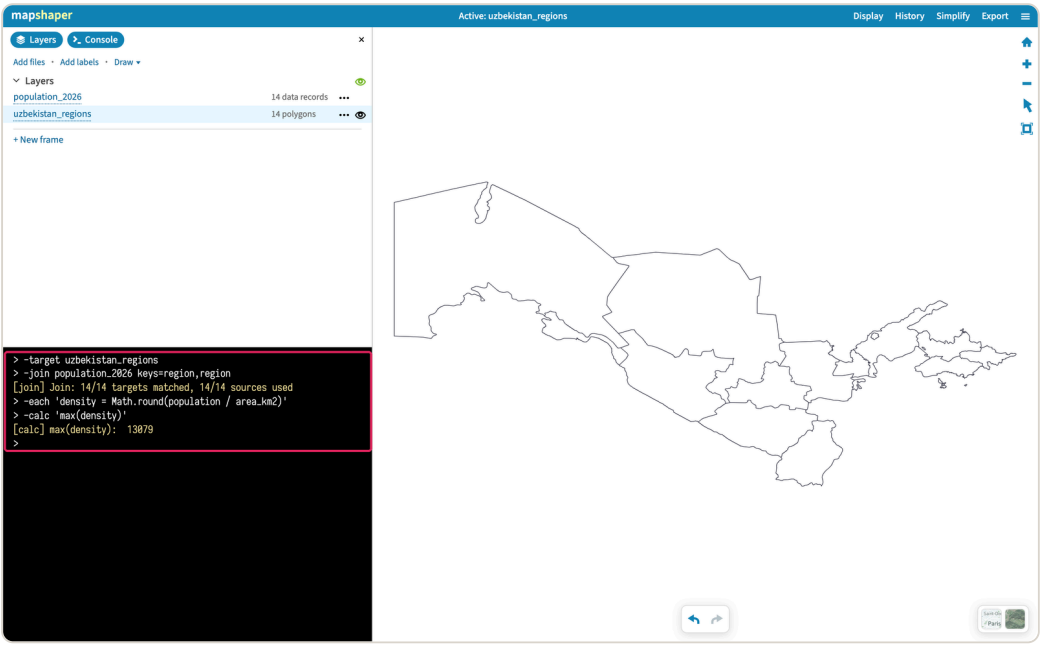


💡 The map turns into a column of empty boxes: Mapshaper is showing the new table layer, which has rows but no shapes. That's normal. The first command in Part 7 switches back to the polygons (or click `uzbekistan_regions` in the list).

🔗 Part 7 · Join and calculate the density

In the console, type these four commands, one by one:

```
-target uzbekistan_regions
-join population_2026 keys=region,region
-each 'density = Math.round(population / area_km2)'
-calc 'max(density)'
```



- `-target` chooses the layer the next commands work on: our polygons.

- `-join` copies the `population` of each row in the table to the polygon with the same `region` name. Look for **14/14 targets matched**: every region found its number. 🎉
- `-each` creates the column `density` = people ÷ km².
- `-calc 'max(density)'` shows the highest density: **13,079 people per km²**, in Tashkent City.

Here is the whole table you just made:

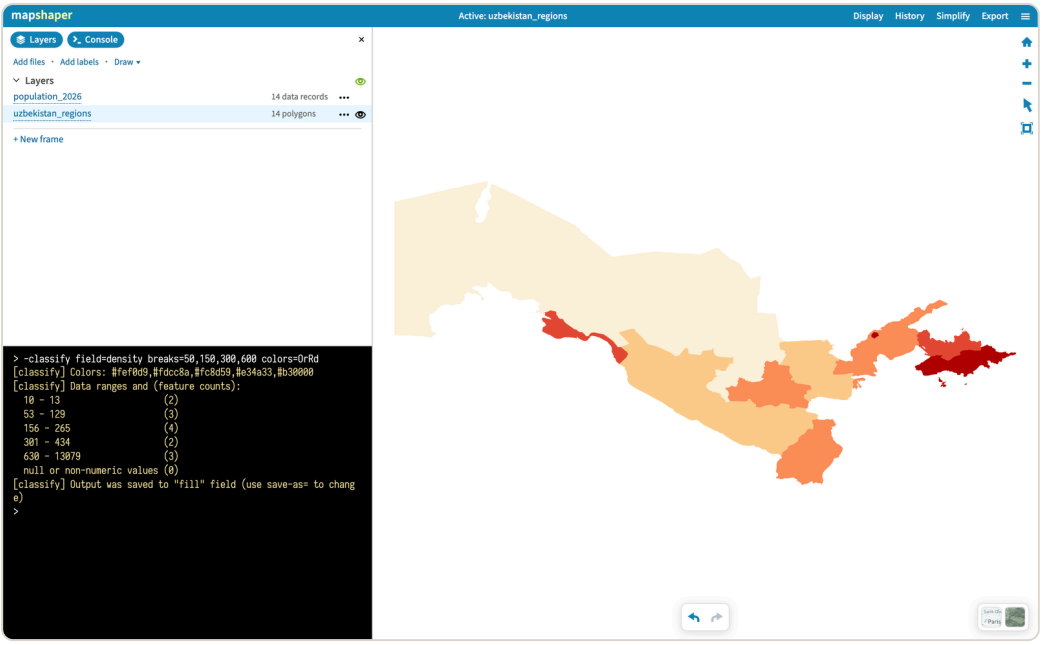
Region	Population	Area (km²)	People per km²
Tashkent City	3,178,100	243	13,079
Andijon	3,521,800	4,079	863
Ferghana	4,223,000	6,701	630
Namangan	3,190,800	7,346	434
Khorezm	2,069,200	6,882	301
Samarkand	4,379,800	16,532	265
Tashkent Region	3,160,700	14,631	216
Sirdaryo	946,300	4,762	199
Surkhandarya	3,011,000	19,269	156
Kashkadarya	3,717,000	28,718	129
Jizzakh	1,566,900	20,789	75
Bukhoro	2,107,000	39,975	53
Karakalpakstan	2,053,200	161,627	13
Navoi	1,111,900	109,615	10

Samarkand has the **most people**, but Andijon is the most **crowded** region: 863 people on every km². Navoi and Karakalpakstan are mostly desert.

The areas are measured from our boundary file, so they differ a little from official figures. Tashkent City's border in this file is older and smaller than today's city, so its density comes out higher than official statistics.

🎨 Part 8 · Colour the polygons (choropleth)

```
-classify field=density breaks=50,150,300,600 colors=OrRd
```



`-classify` puts every region into a colour group:

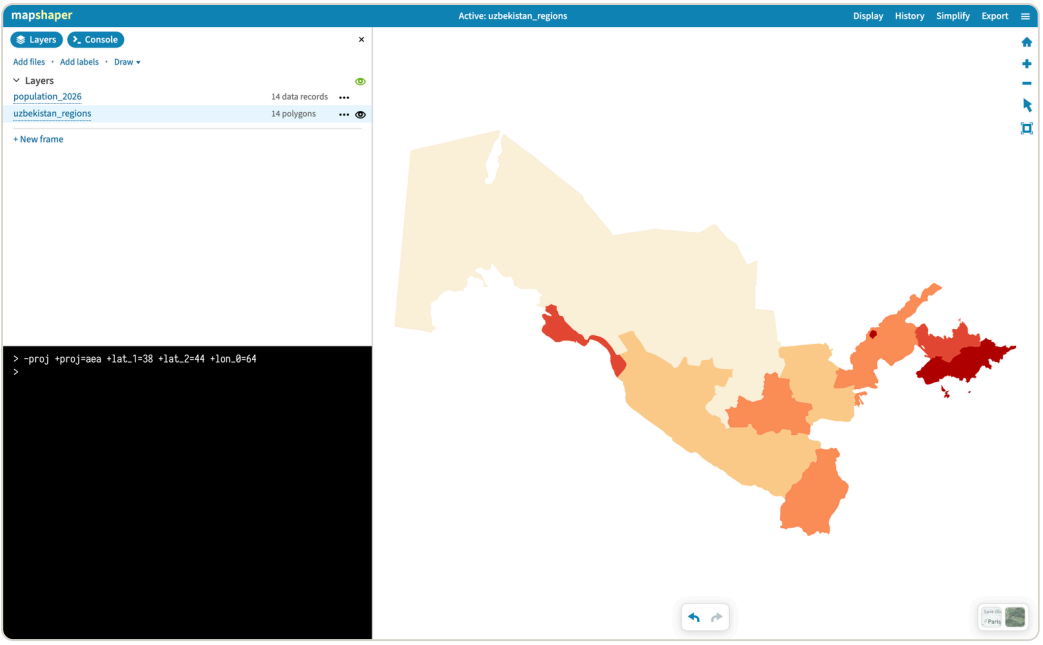
- `field=density` is the number to colour by.
- `breaks=50,150,300,600` are the borders between groups: under 50, 50–150, 150–300, 300–600, and over 600 people per km².
- `colors=OrRd` is the palette (orange to red).

Try other palettes! Replace `OrRd` with `PuRd`, `BuPu`, `YlGnBu`, `PuBuGn`, `Greens` or `Purples` and run the command again. You can also write your own colours: `colors=#FDE2E4,#FAB5C1,#F48AA3,#D65A85,#8E2C5C`.

🌐 Part 9 · A better shape for Uzbekistan (projection)

The map so far uses longitude and latitude directly, which stretches the country sideways. A **projection** draws the round Earth on a flat page. This one keeps the true sizes of areas:

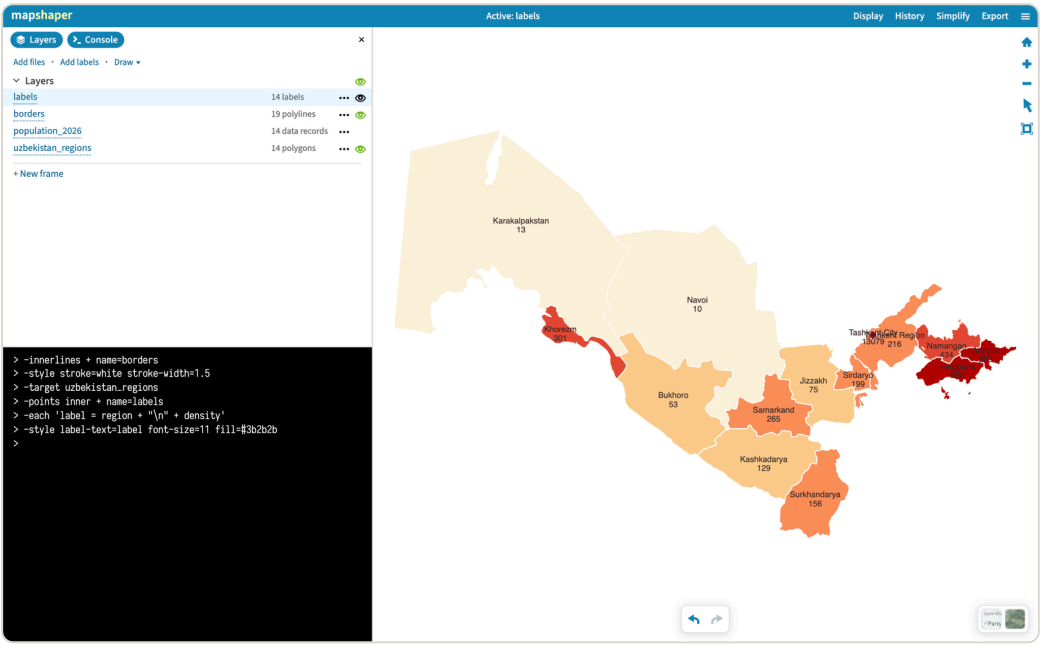
```
-proj +proj=aea +lat_1=38 +lat_2=44 +lon_0=64
```



Uzbekistan now looks the way it does on a globe.

Part 10 · Borders and labels

```
-innerlines + name=borders
-style stroke=white stroke-width=1.5
-target uzbekistan_regions
-points inner + name=labels
-each 'label = region + "\n" + density'
-style label-text=label font-size=11 fill=#3b2b2b
```



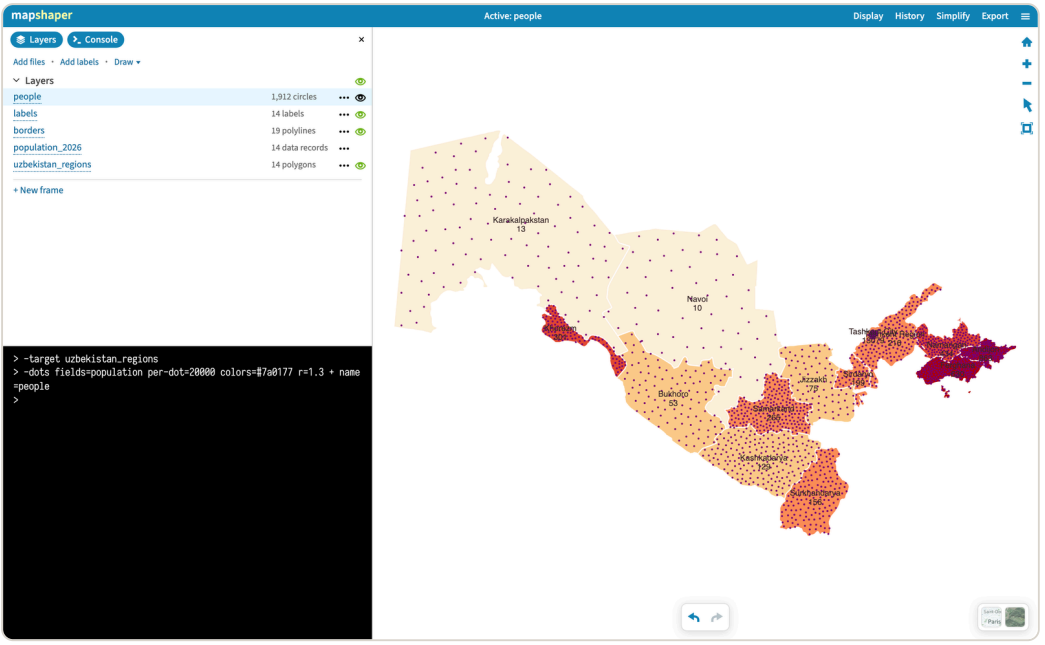
- `-innerlines` creates a new layer with only the **shared borders** between regions (the `+` means "keep the original layer too"). We colour them white.
- `-points inner` puts one point **inside** each polygon, a good place for its label.

- The label is the region name, a new line (`\n`) and the density.

⚫ Part 11 · A dot map: every dot is people

A choropleth shows *density*. A **dot map** shows *how many*: each dot is 20,000 people, scattered randomly inside its region.

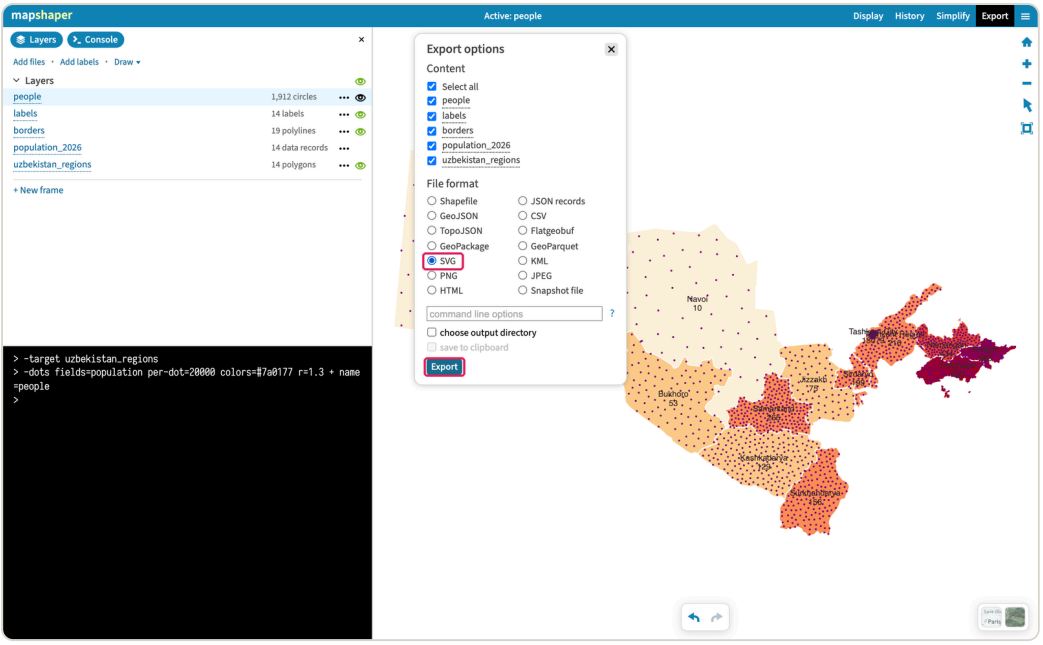
```
-target uzbekistan_regions
-dots fields=population per-dot=20000 colors=#7a0177 r=1.3 + name=people
```



The Ferghana Valley fills up with dots, and the desert stays almost empty. To hide a layer, click its 👁 eye in the **Layers** list.

💾 Part 12 · Export your map

1. Click **Export** (top right).
2. Under **Content**, tick the layers you want in the picture (for the choropleth: `uzbekistan_regions`, `borders`, `labels`).
3. Choose **SVG** (sharp at any size, best for Canva or Figma) or **PNG** (a normal picture), then click **Export**.



The file downloads as `output.svg` (or `.png`). Open it in **Canva**, **Figma** or **Inkscape** to add a **title**, a **legend** for the colours, and the data source.

🌈 Make it yours: ideas

- **Your palette:** change `colors=` in Part 8, or use the night look: a dark background in Canva with `colors=PuBuGn`.
- **More dots:** the dot map at the top uses `per-dot=10000 r=2`. Try fewer, bigger dots with `per-dot=50000 r=2.5`.
- **Two maps side by side:** population (dots) and density (colours) tell different stories. Post both!
- **Only your region:** before Part 8, type `-filter 'region == "Samarkand"'` to keep just one polygon (`-filter` deletes the others, so use ↶ undo to bring them back).
- **Challenge:** Tashkent City's 12 districts are in `Boundaries/Tashkent_boundaries.geojson`. Can you find their populations on stat.uz and make a district map?

💾 Share your map

1. Make a folder named after **your map's topic**, e.g. `Crowded Ferghana Valley`. Put your map inside, plus a short `README.md` with **your name**, the data (stat.uz, simplemaps.com) and the tool (Mapshaper).
2. Upload the folder to **`2026/Day 03 Polygons/`** in this repository. The Quick Start in the main README shows how, with no coding needed.
3. Post it with **#30DayMapChallenge** and credit the **Statistics Agency of Uzbekistan** for the population data.

🆘 Something went wrong?

What you see	What to do
`[join] Missing data layer [population_2026]`	The CSV isn't loaded yet. Do Part 6 first, and check the layer name in the **Layers** list.
`ReferenceError ... population is not defined`	The join didn't happen. Run `-target uzbekistan_regions` and the `-join` command again.
`ReferenceError ... area_km2 is not defined`	You skipped Part 5. Run the `-each 'area_km2 = ...'` command.
`[join] Join: 13/14 targets matched`	A region name in the CSV is spelled differently. Don't edit the files, or copy them again from Part 1.
`The i command cannot be run` (or similar)	You typed a word without the dash. Commands start with `-`, e.g. `-info`.
The map shows empty boxes	The table layer is active. Click `uzbekistan_regions` in the **Layers** list.
A label or point is missing	The slider in Part 4 isn't back at 100%. Click **Simplify** and drag the slider all the way left.
Everything is broken	Reload the page and start again from Part 2. It only takes a few minutes.

📎 Credits

- Population on 1 January 2026: Statistics Agency of Uzbekistan (stat.uz).
- Region boundaries: simplemaps.com, from `Boundaries` in this repository.
- Tool: Mapshaper by Matthew Bloch, open source (GitHub).